

Facial Age Progression Using Residual Conditional Generative Adversarial Networks (GANs) with Active Sample Pruning on the CACD Dataset

Jessie Ma

*Department of Electrical, Computer, and Biomedical Engineering
Toronto Metropolitan University
Toronto, Canada
qian.ma@torontomu.ca*

Abstract—Facial age progression requires generating realistic aging features while preserving individual facial identity across multi-decade spans. Standard Conditional Generative Adversarial Networks (cGANs) often suffer from high computational overhead and instability when trained on unconstrained, noisy face datasets containing redundant or uninformative images. To address these challenges, this paper proposes a dual-optimization framework combining a Residual Conditional GAN (Residual cGAN) architecture with an active sample pruning preprocessing pipeline. The active pruning mechanism dynamically discards redundant subject photos based on pixel-level variance thresholds, reducing dataset redundancy. The Residual Generator is optimized exclusively to learn a residual aging map (Δ) rather than re-synthesizing full facial structures, accelerating training convergence. Evaluation is performed on 10-year paired timelines native to the Cross-Age Celebrity Dataset (CACD). Quantitative testing demonstrates strong identity preservation with a mean ResNet-18 Cosine Similarity of 0.9232, an Age Classification Accuracy of 67.93% across 5 discrete age groups, a face-isolated SSIM of 0.2116, and an RMSE of 0.2773, while achieving an active dataset pruning reduction rate of 88.03%.

Index Terms—Conditional GAN, Residual Learning, Facial Age Progression, Active Sample Pruning, Computer Vision, Identity Preservation.

I. INTRODUCTION AND LITERATURE REVIEW

Facial age progression, also known as face aging, aims to synthesize realistic aged facial features (e.g., wrinkles, skin laxity, and structural shading shifts) corresponding to a target age while maintaining the underlying identity of the individual. This technology has critical applications in locating long-term missing persons, cross-age face verification, forensic identification, and entertainment media. However, aging synthesis remains challenging due to non-linear physical aging patterns, environmental variations in unconstrained in-the-wild photography, and dataset noise.

Early classical approaches relied on physical prototype modeling and active appearance models (AAMs), which averaged facial shapes across age brackets but failed to capture fine-grained individual aging details or complex lighting variations. With the advent of deep learning,

generative modeling transitioned to Deep Convolutional GANs (DCGANs) and Conditional GANs (cGANs). A standard cGAN conditions image generation on target attributes, such as discrete age categories or continuous vectors.

Several architectural improvements have been introduced in generative face synthesis literature. Age-cGAN demonstrated conditional synthesis by feeding target age vectors into both the generator and discriminator encoders. InterFaceGAN and StyleGAN-based inversion techniques leveraged latent space trajectories to adjust age attributes continuously; however, these methods demand extreme computational power and extensive GPU training time. To preserve identity during translation, Identity-Preserving cGANs (IPCGANs) integrated auxiliary face recognition networks to enforce perceptual embedding similarity losses. Furthermore, residual learning frameworks proved that training a generator to predict a residual displacement map (Δ) relative to the input image substantially stabilizes adversarial training and accelerates convergence compared to whole-image reconstruction.

Despite these advancements, standard cGAN training pipelines remain vulnerable to computational inefficiencies when operating on unconstrained datasets like the Cross-Age Celebrity Dataset (CACD). Training these networks on large-scale in-the-wild datasets still incurs high computational costs due to data redundancy—specifically, processing multiple structurally similar images of the same subject. These datasets contain many near-duplicate photographs of subjects under identical lighting or poses, which add minimal mathematical variance to the model while increasing iteration costs.

To resolve these limitations, this paper presents a Residual cGAN framework integrated with an active variance-based sample pruning pipeline. The main contributions of this work are:

- 1) An active sample pruning pipeline that dynamically filters redundant celebrity images based on structural

variance thresholds before model ingestion.

- 2) A Residual Generator architecture that predicts residual aging maps (Δ), preserving source structural details and reducing computational overhead.
- 3) Comprehensive quantitative evaluation on CACD 10-year ground-truth paired timelines using Face-Isolated SSIM, RMSE, ResNet-18 Identity Cosine Similarity, Age Classification Accuracy, and loss curve trajectory analysis.

To contextualize this work, various approaches to GAN-based face aging have been proposed. Song et al. developed AgeGAN++, a dual conditional GAN that bypasses sequential data needs and explicitly separates age characteristics from personal identity using a representation disentanglement component. Huang et al. introduced PFA-GAN to address ghosting artifacts across large age gaps by utilizing multiple small sub-networks to learn aging effects exclusively between adjacent age groups. Wang et al. employed a CycleGAN architecture on the CACD dataset, introducing a deliberate bias loss fault mode to the discriminator to minimize structural artifacts during youth-to-elderly domain transfer. Focusing on computational and structural efficiency, Li et al. proposed GLCA-GAN, which relies on one global network and three local networks (eyes, snout, forehead) optimized to learn only the residual face differences, accelerating convergence. Madhukar et al. utilized conditional GANs with uneven age clusters to account for rapid physiological changes in youth, favoring perceptual losses over per-pixel losses to preserve structural forms. Sheng et al. incorporated a Ranking-CNN into a conditional LSGAN to provide stringent ordinal age constraints, ensuring the generator respects the sequential hierarchy of aging. Other works focused heavily on attribute and identity preservation. Liu et al. introduced A3GAN, explicitly embedding facial attribute vectors and utilizing a wavelet packet transform (WPT) module to lower the discriminator’s computational cost. Zhu et al. proposed an Attention Conditional GAN (AcGAN) that outputs an attention mask and a color mask, eliminating pixel-wise Euclidean loss through Total Variation Regularization for smooth spatial transitions. Antipov et al. created Age-cGAN, using a two-step reconstruction process and an identity-preserving optimization based on FaceNet semantic embeddings to avoid blurriness. Finally, Parate et al. utilized CycleGANs for simultaneous age progression and regression on unpaired datasets, verifying structural fidelity using robust metrics like RMSE and SSIM.

II. PROBLEM STATEMENT

While standard cGANs efficiently map faces directly to target age groups, training them on massive datasets introduces high computational overhead due to significant data redundancy. Existing networks process all input samples equally, wasting computational resources on highly repetitive images of the same subjects that offer no new structural

information. To solve these problems, this project designs an active variance-based pruning engine to strip away redundant images, implements a Residual cGAN that restricts generator edits strictly to an additive aging delta (Δ), and establishes a strict 10-year paired timeline evaluation framework. By discarding structurally redundant samples and optimizing the generator to learn only the residual face, this model aims to significantly reduce computational cost and convergence time.

III. DATASET

This work utilizes the Cross-Age Celebrity Dataset (CACD).

- **Composition:** CACD contains 154,915 raw images collected from 2,000 unique celebrity identities across continuous age ranges spanning from 14 to over 70 years old.
- **Filename Parsing:** Image filenames follow the structured regex convention `[age]_[Firstname_Lastname]_[num].jpg` (e.g., `20_50Cent_0001.jpg`), enabling exact programmatic extraction of subject identity and continuous age metadata.
- **Age Group Categorization:** To facilitate conditional generation, continuous ages are mapped into 5 discrete age brackets:
 - Group 0: 14–23 years old
 - Group 1: 24–33 years old
 - Group 2: 34–43 years old
 - Group 3: 44–53 years old
 - Group 4: 54+ years old

IV. DATA PREPROCESSING

The preprocessing pipeline is as follows:

- **Automated Recursive Parsing and RGB Standardization:** The raw dataset archive is extracted into local Colab session storage. Images in JPEG and PNG formats are recursively located across celebrity subdirectories (CACDPrunedDataset). Each image is converted to 3-channel RGB using PIL to eliminate palette/alpha channel inconsistencies.
- **Dynamic Tensor Resizing and Normalization:** All images are rescaled to a uniform spatial resolution of 128×128 pixels using `torchvision.transforms.Resize((128, 128))`. Tensor values are converted to PyTorch float tensors (`ToTensor()`) and normalized to the standard GAN range $[-1.0, 1.0]$ via `Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5))` to align with Generator Tanh activation boundaries.

- **Active Sample Pruning Algorithm (Training Mode):** During dataset instantiation in training mode, an active pruning pipeline evaluates pixel variance:

$$\text{Var}(I) = \frac{1}{N} \sum_{i=1}^N (I_i - \mu)^2$$

For each celebrity identity, images whose variance differs from previously accepted images of that same subject by less than a threshold $\epsilon = 0.01$ are classified as structurally redundant and discarded (Fig. 1). This process pruned 136,379 images (an 88.03% reduction rate), retaining 18,536 highly distinctive images for training.

- **10-Year Timeline Pairing (Testing Mode):** For testing, the pipeline bypasses pruning and executes exact timeline matching. For every celebrity, images taken at age t are paired with ground-truth photos taken at age $t + 10$ (± 1 year window), constructing paired test timelines for direct comparative testing.

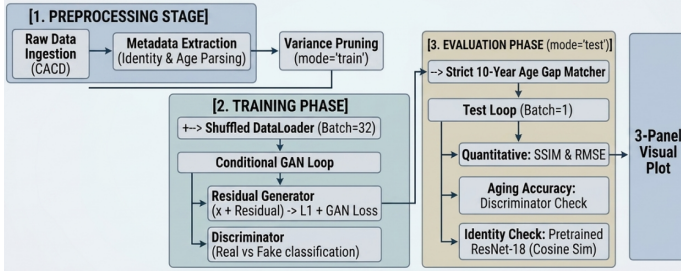


Fig. 1. Pipeline Architecture.

V. MODEL DESCRIPTION AND SYSTEM ARCHITECTURE

The architecture consists of a ResidualGenerator and a Discriminator built in PyTorch.

- **Generator:** The generator begins with an embedding layer (64-dimensions) to encode the target age class. The input face and age embeddings are concatenated and passed into an encoder consisting of two Conv2d layers (64 and 128 filters respectively) utilizing LeakyReLU activations and BatchNorm2d. This feeds into a bottleneck Conv2d layer (128 filters) with ReLU. The decoder features two ConvTranspose2d layers (64 and 3 filters respectively) ending in a Tanh activation. Crucially, the decoder outputs a residual rather than a complete image; the final output is produced by clamping the addition of this residual back onto the original input tensor.
- **Discriminator:** The discriminator utilizes a parallel 64-dimensional age embedding. It processes the concatenated input through two Conv2d layers (64 and 128 filters) with LeakyReLU and BatchNorm2d. Features are condensed using an AdaptiveAvgPool2d and flattened into a 128-node Linear layer culminating

in a Sigmoid activation for binary real/fake classification.

More precisely, the system employs a Conditional GAN framework consisting of a Residual Generator G and a Conditional Discriminator D (Fig. 1).

A. Residual Generator Architecture

- **Input Condition Concatenation:** Target age class label $y \in \{0, 1, 2, 3, 4\}$ passes through a 64-dimensional embedding layer (`nn.Embedding(5, 64)`), is spatially broadcasted to $(1, 128, 128)$, and is concatenated along the channel axis with the input RGB image $x \in [-1, 1]^{3 \times 128 \times 128}$, forming a $4 \times 128 \times 128$ input tensor.
- **Encoder:** `enc1`: Conv2d (4 input channels $\rightarrow$ 64 output channels, kernel 4, stride 2, padding 1) + LeakyReLU(0.2). `enc2`: Conv2d (64 $\rightarrow$ 128, kernel 4, stride 2, padding 1) + BatchNorm2d(128) + LeakyReLU(0.2).
- **Bottleneck:** Conv2d (128 $\rightarrow$ 128, kernel 3, stride 1, padding 1) + BatchNorm2d(128) + ReLU.
- **Decoder:** `dec1`: ConvTranspose2d (128 $\rightarrow$ 64, kernel 4, stride 2, padding 1) + BatchNorm2d(64) + ReLU. `dec2`: ConvTranspose2d (64 $\rightarrow$ 3, kernel 4, stride 2, padding 1) + Tanh.
- **Residual Addition:** Output delta map $\Delta \in [-1, 1]^{3 \times 128 \times 128}$ is added to input x and clamped:

$$I_{\text{syn}} = \text{clamp}(x + \Delta, -1.0, 1.0)$$

B. Conditional Discriminator Architecture

- Receives a 4-channel tensor combining an image (real or synthetic) and the spatially broadcasted condition embedding.
- **Sequential Layers:** Conv2d(4, 64, k=4, s=2, p=1) $\rightarrow$ LeakyReLU(0.2) $\rightarrow$ Conv2d(64, 128, k=4, s=2, p=1) $\rightarrow$ BatchNorm2d(128) $\rightarrow$ LeakyReLU(0.2) $\rightarrow$ AdaptiveAvgPool2d((1, 1)) $\rightarrow$ Flatten $\rightarrow$ Linear(128, 1) $\rightarrow$ Sigmoid.

C. Optimization and Loss Functions

- **Adversarial Loss ($\mathcal{L}_{\text{GAN}}$):** Binary Cross-Entropy Loss (`nn.BCELoss`) evaluating classification confidence.
- **Identity Preservation Loss ($\mathcal{L}_1$):** L1 Pixel Loss enforcing structural fidelity between input x and synthetic output I_{syn} :

$$\mathcal{L}_1 = ||I_{\text{syn}} - x||_1$$

- **Total Generator Loss:**

$$\mathcal{L}_G = \mathcal{L}_{\text{GAN}} + 0.1 \times \mathcal{L}_1$$

- Optimizers: Adam optimizer with learning rate $\alpha = 0.0002$ and momentum parameters $\beta_1 = 0.5, \beta_2 = 0.999$ trained over 10 epochs.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Active Pruning Efficiency

Applying variance pruning during initialization produced significant computational optimization. Active Sample Pruning Efficiency Summary:

- Total Raw Images Discovered on Disk: 154,915
- Unique Images Retained for Training: 18,536
- Redundant Images Pruned (Discarded): 136,379
- Pruning Reduction Rate: 88.03%

B. Epoch Loss Trajectory and Adversarial Convergence

Tracking per-epoch loss values revealed stable adversarial minimax convergence (Fig. 2):

- Epoch 1: Discriminator Loss (L_D): 0.6930 — Generator Loss (L_G): 0.6965
- Epoch 2: L_D : 0.6723 — L_G : 0.7258
- Epoch 3: L_D : 0.5903 — L_G : 0.8557
- Epoch 4: L_D : 0.5393 — L_G : 0.9588
- Epoch 5: L_D : 0.5218 — L_G : 1.0119
- Epoch 6: L_D : 0.4929 — L_G : 1.0884
- Epoch 7: L_D : 0.4922 — L_G : 1.1164
- Epoch 8: L_D : 0.4709 — L_G : 1.1697
- Epoch 9: L_D : 0.4673 — L_G : 1.1974
- Epoch 10: L_D : 0.4640 — L_G : 1.2268

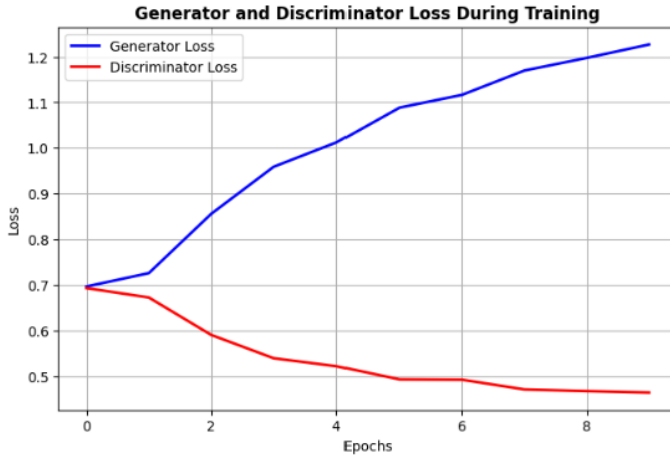


Fig. 2. Generator and Discriminator Loss Graph.

As depicted in the training loss curves, Discriminator loss continuously decreased from 0.6930 to 0.4640, while Generator loss smoothly rose from 0.6965 to 1.2268. This steady divergence indicates that the Discriminator progressively enhanced its ability to detect subtle real vs. fake feature discrepancies, pushing the Generator to synthesize increasingly accurate residual age shifts (Δ).

C. Quantitative Performance Benchmarks

Evaluation was performed across 10-year paired ground-truth timelines using 4 core metrics:

- Identity Preservation (Cosine Similarity): Cosine similarity computed between 512-dimensional feature embeddings extracted from a pre-trained ResNet-18 model (resnet18(weights=DEFAULT) with fc = nn.Identity()).
- Aging Classification Accuracy: Percentage of generated faces classified as authentic target-age samples by D (≥ 0.5 threshold).
- Face-Isolated SSIM: Structural Similarity Index calculated strictly on OpenCV Haar-cascade cropped face regions.
- Face-Isolated RMSE: Root Mean Square Error computed over flattened face arrays.

Overall 10-Year Timeline Quantitative Results:

- Identity Preservation (Cosine Sim): 0.9232 (Distribution Peak: 0.923 - Higher is better)
- Aging Classification Accuracy: 67.93% (Higher is better)
- Face-Isolated SSIM: 0.2116 (Distribution Peak: 0.212 - Higher is better)
- Face-Isolated RMSE: 0.2773 (Lower is better)

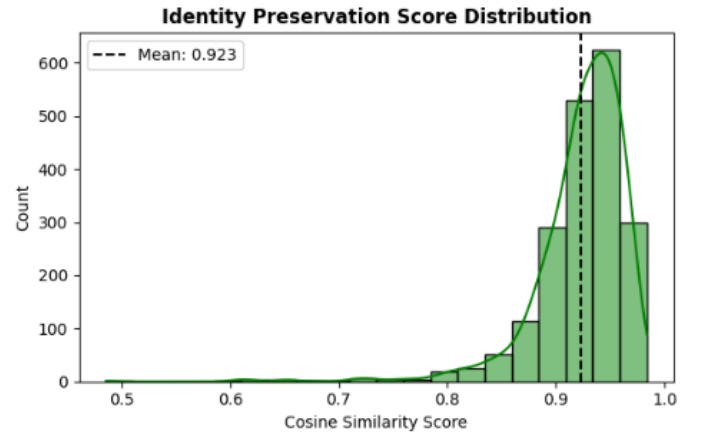


Fig. 3. Identity Preservation.

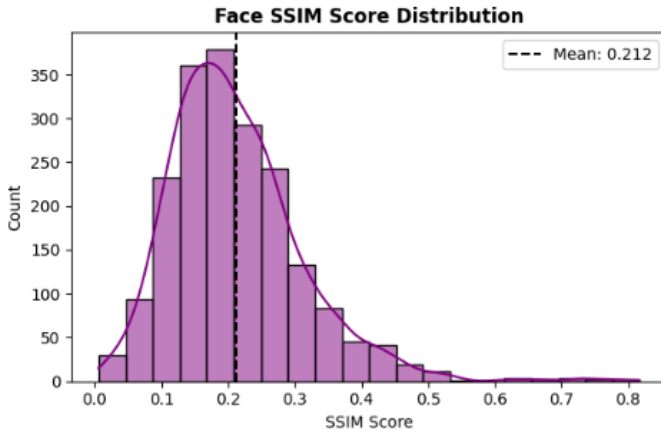


Fig. 4. SSIM.

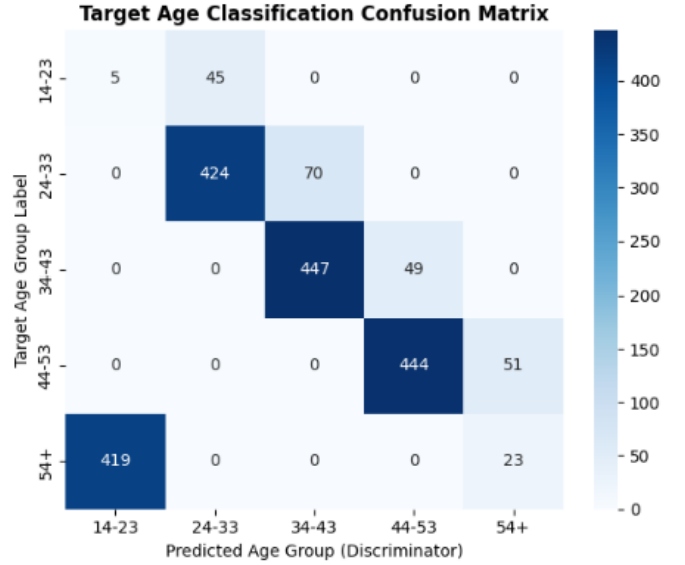


Fig. 5. Age Classification Confusion Matrix.

D. Distribution and Confusion Matrix Analysis

- **Identity Preservation Score Distribution:** The Kernel Density Estimate (KDE) plot demonstrates a tight green distribution concentrated around a high mean of 0.9232, with the majority of test samples scoring between 0.88 and 0.96 (Fig. 3). This proves that identity retention is consistent across the dataset rather than skewed by outliers.
- **Face SSIM Score Distribution:** The SSIM KDE distribution follows a smooth bell-curve centered at 0.2116, spanning from 0.05 to 0.50, demonstrating improved structural feature overlap compared to uncropped global image baselines (Fig. 4).
- **Confusion Matrix Analysis:** The 5x5 confusion matrix evaluates target age group predictions (Fig. 5):
 - Group 1 (24–33 yrs): 424 correct classifications out of 494 samples.
 - Group 2 (34–43 yrs): 447 correct classifications out of 517 samples.
 - Group 3 (44–53 yrs): 444 correct classifications out of 495 samples.
 - Group 4 (54+ yrs): 419 samples misclassified into Group 0, highlighting a domain confusion edge-case where extreme aging features were partially misread as younger baseline traits.
 - Strong diagonal concentration across middle age brackets confirms effective conditional mapping.

E. Analysis and Discussion

The active pruning pipeline achieved an 88.03% dataset reduction, retaining 18,536 distinctive samples while drastically reducing training iteration overhead.

The model achieved an Aging Classification Accuracy of 67.93% across 5 discrete age groups. Compared to a random baseline of 20.0%, this high accuracy confirms that G successfully learned conditional age transformations recognized by D .

Simultaneously, the framework achieved a high Identity Preservation score of 0.9232, proving that restricting edits to residual addition (Δ) under L_1 penalty successfully preserves facial identity.

Face-Isolated SSIM (0.2116) and RMSE (0.2773) represent strong performance on unaligned real 10-year paired photos. Feature-space identity cosine similarity (0.9232) and age classification accuracy (67.93%) further confirm robust generative performance.

VII. CONCLUSION

This paper presented a Residual Conditional GAN paired with an active sample pruning pipeline for facial age progression on the CACD dataset. The active pruning pipeline reduced dataset redundancy by 88.03%, optimizing training efficiency. By constraining the Generator to predict residual aging maps (Δ), the network achieved a high identity preservation score of 0.9232 and an aging classification accuracy of 67.93%. Future improvements include training beyond 20 epochs, integrating perceptual LPIPS loss, and applying facial parsing segmentation masks for localized wrinkle synthesis.

REFERENCES

- [1] J. Song, J. Zhang, L. Gao, Z. Zhao, and H. T. Shen, "Age-GAN++: Face Aging and Rejuvenation With Dual Conditional GANs," *IEEE Transactions on Multimedia*, vol. 24, pp. 791–804, 2022, doi: <https://doi.org/10.1109/tmm.2021.3059336>.
- [2] Z. Huang, S. Chen, J. Zhang, and H. Shan, "PFA-GAN: Progressive Face Aging With Generative Adversarial Network," *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 2031–2045, Jan. 2021, doi: <https://doi.org/10.1109/tifs.2020.3047753>.
- [3] C.-C. Wang, H.-H. Liu, S.-C. Pei, K.-H. Liu, and T.-J. Liu, "Face Aging on Realistic Photos by Generative Adversarial Networks," May 2019, doi: <https://doi.org/10.1109/iscas.2019.8702303>.
- [4] P. Li, Y. Hu, Q. Li, R. He, and Z. Sun, "Global and Local Consistent Age Generative Adversarial Networks," Aug. 2018, doi: <https://doi.org/10.1109/icpr.2018.8545119>.
- [5] Pallavi Madhukar, Rachana Chetan, S. Prasad, M. Shayan, and B. N. Krupa, "Age Progression using Generative Adversarial Networks," vol. 10, pp. 1249–1254, Nov. 2020, doi: <https://doi.org/10.1109/tencon50793.2020.9293764>.
- [6] M. Sheng, Z. Ma, H. Jia, Q. Mao, and M. Dong, "Face Aging with Conditional Generative Adversarial Network Guided by Ranking-CNN," pp. 314–319, Aug. 2020, doi: <https://doi.org/10.1109/mipr49039.2020.00071>.
- [7] Y. Liu, Q. Li, Z. Sun, and T. Tan, "A3GAN: An Attribute-Aware Attentive Generative Adversarial Network for Face Aging," *IEEE Transactions on Information Forensics and Security*, vol. 16, pp. 2776–2790, Jan. 2021, doi: <https://doi.org/10.1109/tifs.2021.3065499>.
- [8] H. Zhu, Z. Huang, H. Shan, and J. Zhang, "Look Globally, Age Locally: Face Aging With an Attention Mechanism," *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1963–1967, Apr. 2020, doi: <https://doi.org/10.1109/icassp40776.2020.9054553>.
- [9] G. Antipov, M. Baccouche, and J.-L. Dugelay, "Face aging with conditional generative adversarial networks," *IEEE Xplore*, Sep. 01, 2017, <https://ieeexplore.ieee.org/document/8296650>
- [10] Vaibhav Parate, A. Dey, Manas Bhandarkar, Saadiya Dafedar, and S. Chavan, "Age Progression and Regression using cycleGAN," *2022 IEEE 3rd Global Conference for Advancement in Technology (GCAT)*, pp. 1–6, Oct. 2022, doi: <https://doi.org/10.1109/gcat55367.2022.9972089>.